CS502 - Fundamentals of Algorithms Solved MCQs from Quiz 3

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Solved by Umair Saulat

In in-place sorting algorithm is one that uses arrays for storage :
An additional array
No additional array (http://www.vuzs.info/)
Both of above may be true according to algorithm
More than 3 arrays of one dimension.

The running time of quick sort depends heavily on the selection of
No of inputs
Arrangement of elements in array
Size o elements
Pivot element (http://www.vuzs.info/)

In stable sorting algorithm
One array is used
In which duplicating elements are not handled.
More then one arrays are required.
Duplicating elements remain in same relative position after sorting. (http://www.vuzs.info/)

Which sorting algorithn is faster :
O(n^2)
O(nlogn)
O(n+k) (http://www.vuzs.info/)
O(n^3)

In Quick sort algorithm,constants hidden in T(n lg n) are
Large
Medium
Not known
Small (http://www.vuzs.info/)

Quick sort is based on divide and conquer paradigm; we divide the problem on base of pivot element and:
There is explicit combine process as well to conquer the solution. (http://www.vuzs.info/)
No work is needed to combine the sub-arrays, the array is already sorted
Merging the subarrays
None of above.

There is relationship between number of back edges and number of cycles in DFS
Both are equal.
Cycles are half of back edges.
Cycles are one fourth of back edges.
There is no relationship between back edges and number of cycle (http://www.vuzs.info/)

You have an adjacency list for G, what is the time complexity to compute Graph transpose G^T ?
(V+E) (http://www.vuzs.info/)
V.E
V
E

You have an adjacency list for G, what is the time complexity to compute Graph transpose G^T.?
?(V + E) (http://www.vuzs.info/)
?(V E)
?(V)
?(V^2)

What is the time complexity to extract a vertex from the priority queue in Prim's algorithm?
log (V) (http://www.vuzs.info/)
V.V
E.E
log (E)

Dijkstra's algorithm :
Has greedy approach to find all shortest paths
Has both greedy and Dynamic approach to find all shortest paths
Has greedy approach to compute single source shortest paths to all other vertices (http://www.vuzs.info/)
Has both greedy and dynamic approach to compute single source shortest paths to all other vertices.

What algorithm technique is used in the implementation of Kruskal solution for the MST?
Greedy Technique (http://www.vuzs.info/)
Divide-and-Conquer Technique
Dynamic Programming Technique
The algorithm combines more than one of the above techniques

Which is true statement in the following.
Kruskal algorithm is multiple source technique for finding MST.
Kruskal's algorithm is used to find minimum spanning tree of a graph, time complexity of this algorithm is O(EV)
Both of above
Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose best Tree edge) when the graph has relatively few edges) (http://www.vuzs.info/)

The relationship between number of back edges and number of cycles in DFS is,
Both are equal
Back edges are half of cycles
Back edges are one quarter of cycles
There is no relationship between no. of edges and cycles (http://www.vuzs.info/)

Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose best tree edge) when the graph has relatively few edges
True (http://www.vuzs.info/)
False

Suppose that a graph G = (V,E) is implemented using adjacency lists. What is the complexity of a breadth-first traversal of G?
O(|V |^2)
O(|V | |E|) (http://www.vuzs.info/)
O(|V |^2|E|)
O(|V | + |E|)

What is generally true of Adjacency List and Adjacency Matrix representations of graphs?
Lists require less space than matrices but take longer to find the weight of an edge (v1,v2)
Lists require less space than matrices and they are faster to find the weight of an edge (v1, v2) (http://www.vuzs.info/)
Lists require more space than matrices and they take longer to find the weight of an edge (v1, v2)
Lists require more space than matrices but are faster to find the weight of an edge (v1, v2)

What general property of the list indicates that the graph has an isolated vertex?
There is Null pointer at the end of list.
The Isolated vertex is not handled in list.
Only one value is entered in the list.
There is at least one null list.

A dense undirected graph is:
A graph in which E = O(V^2) (http://www.vuzs.info/)
A graph in which E = O(V)
A graph in which E = O(log V)
All items above may be used to characterize a dense undirected graph

In digraph G=(V,E) ;G has cycle if and only if
The DFS forest has forward edge.
The DFS forest has back edge (http://www.vuzs.info/)
The DFS forest has both back and forward edge
BFS forest has forward edge

Back edge is:
(u, v) where v is an ancestor of u in the tree. (http://www.vuzs.info/)
(u,v) where u is an ancesstor of v in the tree.
(u, v) where v is an predcessor of u in the tree.
None of above

Using ASCII standard the string "abacdaacacwe" will be encoded with _____ bits
64
128 (http://www.vuzs.info/)
96
120

Cross edge is :
(u, v) where u and v are not ancestor of one another
(u, v) where u is ancesstor of v and v is not descendent of u.
(u, v) where u and v are not ancestor or descendent of one another (http://www.vuzs.info/)
(u, v) where u and v are either ancestor or descendent of one another.

Which statement is true?
If a dynamic-programming problem satisfies the optimal-substructure property, then a locally optimal solution is globally optimal.
If a greedy choice property satisfies the optimal-substructure property, then a locally optimal solution is globally optimal.
Both of above (http://www.vuzs.info/)
None of above

If you find yourself in maze the better traversel approach will bEA dense undirected graph is:
A graph in which E = O(V^2) (http://www.vuzs.info/)
A graph in which E = O(V)
A graph in which E = O(log V)
All items above may be used to characterize a dense undirected graph

Which is true statement.
Breadth first search is shortest path algorithm that works on un-weighted graphs (http://www.vuzs.info/)
Depth first search is shortest path algorithm that works on un-weighted graphs.
Both of above are true.
None of above are true.

Forward edge is:
(u, v) where u is a proper descendent of v in the tree.
(u, v) where v is a proper descendent of u in the tree. (http://www.vuzs.info/)
(u, v) where v is a proper ancesstor of u in the tree.
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BFS
BFS and DFS both are valid (http://www.vuzs.info/)
Level order
DFS

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True (http://www.vuzs.info/)
False

A digraph is strongly connected under what condition?
A digraph is strongly connected if for every pair of vertices u, v e V, u can reach v .
A digraph is strongly connected if for every pair of vertices u, v e V, u can reach v and vice versa. (http://www.vuzs.info/)
A digraph is strongly connected if for at least one pair of vertex u, v e V, u can reach v and vice versa.
A digraph is strongly connected if at least one third pair of vertices u, v e V, u can reach v and vice versa.

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